The Ark AI

Bitcoin Lightning AI Service Platform

Software Documentation for Copyright Registration

Copyright Work Title:	The Ark AI - Bitcoin Lightning AI Service Platform
Software Version:	Version 1.0.0
Software Developer:	[Author Name]
Software Owner:	[Owner Name]
Date of Creation:	February 20, 2026
Date of Publication:	February 20, 2026
Primary URL:	https://arknode.ai
Secondary URL:	https://thenode.it.com
Category:	Literary Work (Computer Program)

1. Abstract

a) Purpose of the Software

The Ark AI is an original software platform that provides artificial intelligence services accessible via Bitcoin Lightning Network micropayments. The platform enables users to access 143+ AI-powered services including text generation, image creation and editing, document generation with automatic PDF export, code assistance, translation across 80+ languages, data analysis, and voice synthesis in multiple African languages - all payable through Bitcoin Lightning invoices or a built-in satoshi wallet system.

The software solves the problem of AI accessibility in developing economies where monthly subscription models (such as ChatGPT Plus at $20/month or Claude Pro at $20/month) are unaffordable for most users. By implementing a pay-per-task model with prices starting at 10 satoshis (approximately $0.007 USD), The Ark AI makes professional AI tools accessible to anyone with a Bitcoin Lightning wallet.

b) Technologies Used

The Ark AI is built using the following technologies:

- Python 3.12 - Core backend programming language
- FastAPI - Asynchronous web framework for the REST API server
- JavaScript (ES6+) - Frontend interface with real-time streaming
- HTML5 / CSS3 - Responsive web interface design
- Server-Sent Events (SSE) - Real-time response streaming protocol
- OpenAI GPT-Image-1.5 API - Image generation and editing
- DeepInfra API (Qwen3-235B) - Text AI model inference
- Bitcoin Lightning Network - Payment protocol (L402)
- Phoenix Lightning Daemon - Lightning node software
- LNbits - Lightning wallet and invoice management
- Nginx - Reverse proxy with SSL/TLS encryption
- fpdf2 - PDF document generation library
- Ubuntu 24.04 LTS - Server operating system
- Docker - Container runtime for LNbits

c) Platforms

The Ark AI runs as a web application accessible via any modern web browser on desktop and mobile devices. The software is deployed on a cloud-based virtual private server (DigitalOcean, Ubuntu 24.04 LTS) and is accessible at the following URLs:

- https://arknode.ai (primary domain)
- https://thenode.it.com (secondary domain)

The platform is designed as a Progressive Web Application (PWA) with service worker support, enabling offline caching and mobile-app-like experience on smartphones. It supports all major browsers including Chrome, Firefox, Safari, and

Edge on desktop, tablet, and mobile platforms.

2. Introduction

The Ark AI is a comprehensive artificial intelligence service platform that democratizes access to AI through Bitcoin Lightning micropayments. Unlike traditional AI platforms that require monthly subscriptions, credit cards, and personal account registration, The Ark AI allows anyone to access professional AI services by simply scanning a QR code with a Bitcoin Lightning wallet.

Key Functionalities

- Text AI Services (100+ types): Summarization, translation, content writing, code generation, research, data analysis, legal document drafting, educational content, email composition, and more
- Image Services (24 types): AI image generation, editing, background removal, colorization, photo restoration, style transfer, upscaling, logo design, avatar creation, and more
- Document Generation: Automatic conversion of AI-generated content into professionally formatted, downloadable PDF documents including privacy policies, contracts, NDAs, business plans, invoices, and more
- Voice Services: Text-to-speech synthesis in multiple languages including African languages (Swahili, Amharic, Zulu, Somali, Afrikaans)
- Translation Services: 80+ language translation with dedicated translation interface supporting text, file, and image translation
- Chama (Group) AI Access: Collaborative AI access system where teams, schools, or communities pool satoshis for shared AI usage
- Lightning Wallet System: Built-in wallet allowing users to maintain satoshi balances for faster repeated access without per-task QR code scanning
- OpenAI-Compatible REST API: Developer API endpoint allowing third-party applications to connect using standard OpenAI client libraries
- Free Trial System: New users receive 3 complimentary AI tasks to experience the platform before paying

How the Software Works

1. A user visits arknode.ai and is presented with the main chat interface showing quick-access buttons for common AI tasks (Write code, Summarize, Translate, Research, Code review, Write email, Swahili voice, Amharic voice).

2. The user types their request in the input field or selects a quick-access button. They can also attach images or files for processing.

3. The system automatically detects the task type from the user's input and routes it to the appropriate AI model and service configuration.

4. For new users (first 3 tasks), the response streams immediately at no cost. For subsequent tasks, a Lightning invoice QR code is displayed showing the price in satoshis.

5. The user scans the QR code with any Bitcoin Lightning wallet (Phoenix, Wallet of Satoshi, Muun, etc.) to pay. Alternatively, users can top up the built-in wallet to skip QR codes on future tasks.

6. Upon payment confirmation, the AI response streams in real-time using Server-Sent Events, providing instant feedback as the response is generated.

7. For document tasks, a downloadable PDF is automatically generated. For image tasks, the generated or edited image is displayed with a download button.

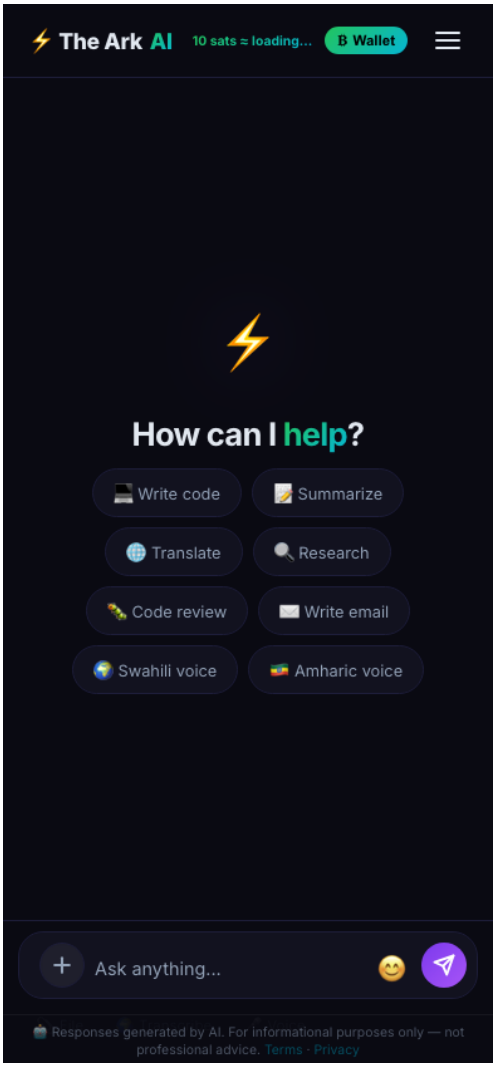
3. Software Documentation

This section provides a sequenced description of the software's key interfaces and functionalities, accompanied by relevant screenshots of the graphical user interface (GUI).

3.1 Main Interface - Home Screen

Upon accessing https://arknode.ai, users are presented with the main chat interface. The home screen features the lightning bolt logo, a greeting message ('How can I help?'), and quick-access buttons for popular AI services. The header displays the current pricing (10 sats), a Bitcoin wallet button, and a hamburger menu for navigation. At the bottom, the chat input field allows users to type requests, attach files, transcribe audio, or use voice input.

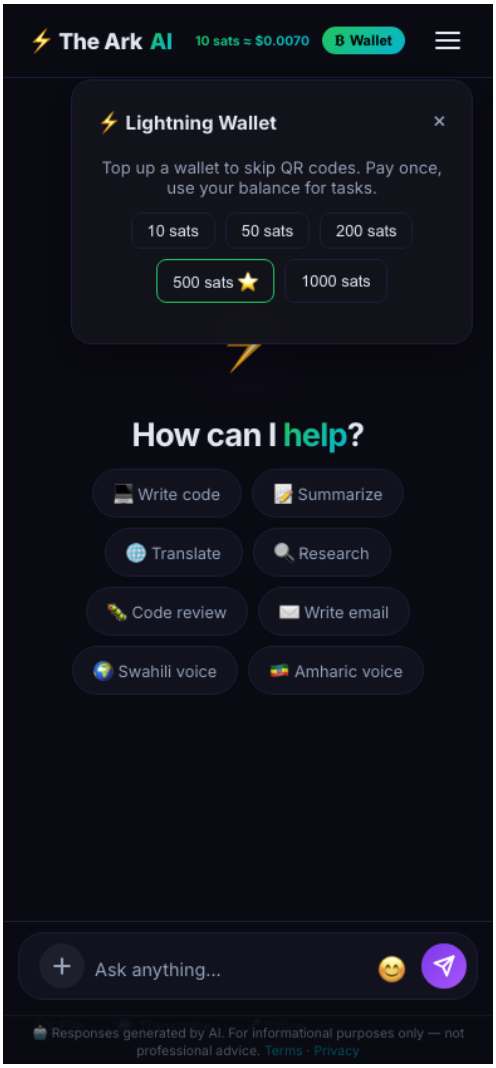
Figure: Main Interface - Home Screen (arknode.ai)



3.2 Lightning Wallet System

Clicking the 'Wallet' button in the header opens the Lightning Wallet popup. Users can top up their wallet balance by selecting preset amounts (10, 50, 200, 500, or 1000 satoshis). The 500 sats option is highlighted as the recommended amount. Once topped up, users can access AI services without scanning individual QR codes for each task, providing a seamless experience similar to a prepaid account.

Figure: Lightning Wallet - Top Up Interface



3.3 Navigation Menu

The hamburger menu provides access to all sections of the platform: Chama AI (group access), Productivity tools, All Services marketplace, Lightning Guide (payment tutorial), Translate (dedicated translation interface), About page, and Contact information. This menu is accessible from every page of the application.

Menu items include: Chama AI, Productivity, All Services, Lightning Guide, Translate, About, and Contact.

3.4 About Page

The About page provides comprehensive information about The Ark AI platform, including pricing tiers, a three-step usage guide (Select a Task, Pay the Invoice, Get Results), unlimited access options, the Chama group AI feature, REST API documentation with code examples, and a Frequently Asked Questions section covering data privacy, payment methods, and service capabilities.

Figure: About Page - Platform Overview and Pricing

3.5 All Services Marketplace

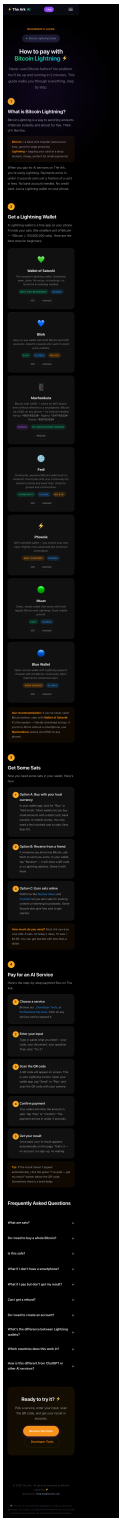
The All Services page displays all 143+ AI services organized by category. Each service shows its name, brief description, and price in satoshis. Categories include: Code & Development, Text & Strategy, Healthcare & Clinical, Education & Research, Real Estate, Business & Productivity, Agent AI Tools, Voice & Image, and more. Users can browse and select any service to begin a task.

Figure: All Services Marketplace - 143+ AI Services

3.6 Lightning Payment Guide

The Lightning Guide page provides a comprehensive tutorial for users new to Bitcoin Lightning payments. It explains what Bitcoin Lightning is, recommends wallet apps (Wallet of Satoshi, Machankura, Phoenix, Blue Wallet), describes how to acquire satoshis, and provides a step-by-step walkthrough of paying for an AI service. The guide includes visual instructions and links to recommended wallet applications.

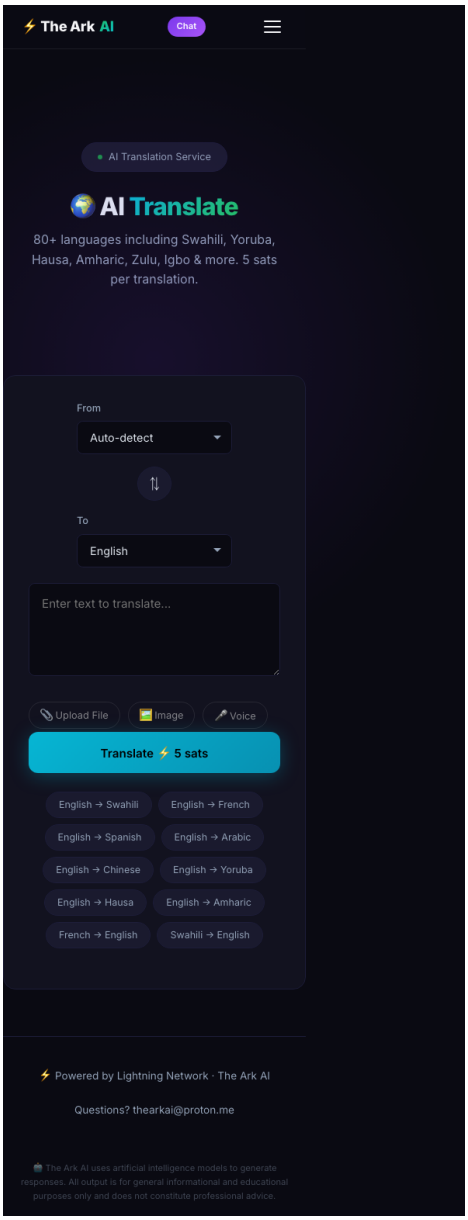
Figure: Lightning Payment Guide - How to Pay with Bitcoin



3.7 Dedicated Translation Interface

The Translate page provides a specialized translation interface supporting 80+ languages with a focus on African languages. The interface features source and target language selectors with auto-detection, a text input area, and options to upload files, images, or use voice input for translation. Quick-access buttons provide one-click translation for popular language pairs including English to Swahili, French, Spanish, Arabic, Chinese, Yoruba, Hausa, and Amharic.

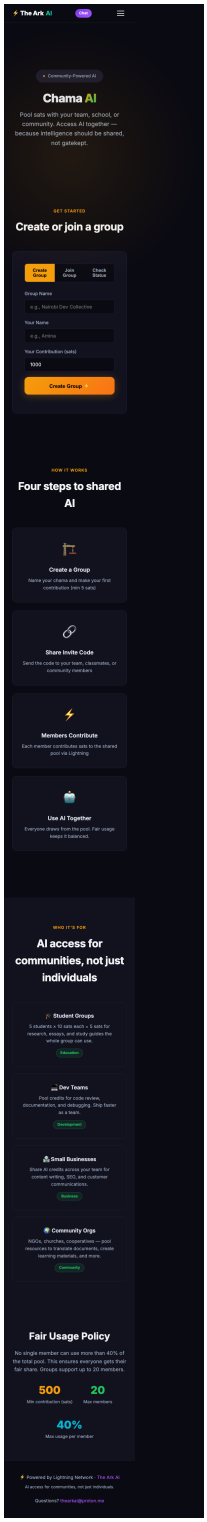
Figure: Translation Interface - 80+ Languages



3.8 Chama - Group AI Access

The Chama (Swahili for 'group') feature enables collaborative AI access for teams, schools, and communities. Users can create a group, share an invite code, and pool satoshi contributions for shared AI usage. The system enforces fair usage (no single member can use more than 40% of the pool). Groups support up to 20 members with a minimum contribution of 500 sats. This feature is designed for student groups, development teams, small businesses, and community organizations who want to share AI resources affordably.

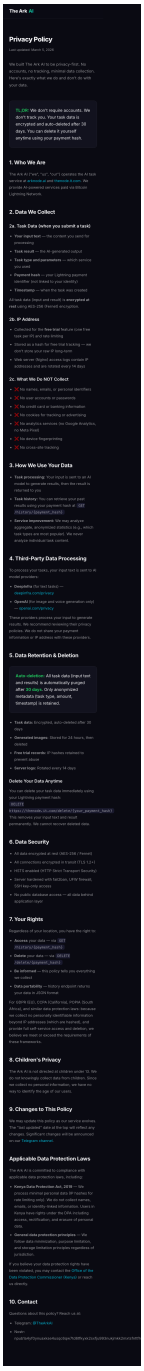
Figure: Chama - Group/Community AI Access



3.9 Privacy Policy

The platform includes a comprehensive privacy policy accessible via the footer link. The policy covers data collection practices, usage, storage, and user rights in compliance with applicable data protection regulations.

Figure: Privacy Policy Page



The Ark AI

Privacy Policy

Last updated: March 1, 2026

We built The Ark AI to be privacy-first. No accounts, no tracking, minimal data collection. Here's exactly what we do and don't do with your data.

TL;DR: We don't require accounts. We don't track you. Your task data is encrypted and auto-deleted after 30 days. You can delete it yourself anytime using your payment hash.

1. Who We Are

The Ark AI ("we", "us", "our") operates the AI Nish service at arkai.nish.al and thearkai.t.nish. We provide AI powered services paid via Bitcoin Lightning Network.

2. Data We Collect

2a. Task Data (when you submit a task)

- Your input text — the content you send for processing.
- Task result — the AI generated output
- Task type and parameters — which service you used
- Payment hash — your Lightning payment identifier (not linked to your identity)
- Timestamps — when the task was created

All task data (input and result) is every priced at rest using AES-256 (Fernet) encryption.

2b. IP Address

- Collected for the free trial feature (one free task per IP) and rate limiting
- Stored as a SHA-256 free trial tracking — we don't store your raw IP long-term
- Web server (Nginx) access logs contain IP addresses and are rotated every 14 days

2c. What We Do NOT Collect

- ❌ No names, emails, or personal identifiers
- ❌ No user accounts or passwords
- ❌ No credit card or banking information
- ❌ No cookies for tracking or advertising
- ❌ No analytics services (no Google Analytics, no Meta Pixel)
- ❌ No device fingerprinting
- ❌ No cross-site tracking

3. How We Use Your Data

- Task processing: Your input is sent to an AI model to generate results. Even the result is sent back to you.
- Task history: You can retrieve your past results using your payment hash at GET /history/{payment_hash}
- Service improvement: We may analyze aggregate, anonymized statistics (e.g., which task types are most popular). We never analyze individual task content.

4. Third-Party Data Processing

To process your tasks, your input text is sent to AI model providers:

- DeepInfra (for text tasks) — deepinfra.com/privacy
- OpenAI (for image and voice generation only) — openai.com/policies/privacy

These providers process your input to generate results. We recommend reviewing their privacy policies. We do not share your payment information or IP address with these providers.

5. Data Retention & Deletion

Auto-deletion: All task data (input text and results) is automatically purged after 30 days. Only anonymized metadata (task type, amount, timestamp) is retained.

- Task data: Encrypted, auto-deleted after 30 days
- Generated images: Stored for 24 hours, then deleted
- Free trial records: IP hashes retained to prevent abuse
- Server logs: Rotated every 14 days

Delete Your Data Anytime

You can delete your task data immediately using your Lightning payment hash.

DELETE https://thearkai.t.nish.al/api/delete/{your_payment_hash}

This removes your input text and result permanently. We cannot recover deleted data.

6. Data Security

- All data encrypted at rest (AES-256 (Fernet))
- All connections encrypted in transit (TLS 1.2+)
- HSTS enabled (HTTP Strict Transport Security)
- Server hardened with fail2ban, UFW firewall, SSH key-only access
- No public database access — all data behind application layer

7. Your Rights

Regardless of your location, you have the right to:

- Access your data — via GET /history/{payment_hash}
- Delete your data — via DELETE /delete/{payment_hash}
- Be informed — this policy tells you everything we collect
- Data portability — history endpoint returns your data in JSON format

For GDPR (EU), CCPA (California), POPIA (South Africa), and similar data protection laws, because we collect no personally identifiable information (beyond IP addresses (which are hashed), and provide full self service access and deletion, we believe we meet or exceed the requirements of these frameworks.

8. Children's Privacy

The Ark AI is not directed at children under 13. We do not knowingly collect data from children. Since we collect no personal information, we have no way to identify the age of our users.

9. Changes to This Policy

We may update this policy as our service evolves. The "last updated" date at the top will reflect any changes. Significant changes will be announced on our Telegram channel.

Applicable Data Protection Laws

The Ark AI is committed to compliance with applicable data protection laws, including:

- Kenya Data Protection Act, 2019 — We process minimal personal data (IP hashes for rate limiting only). We do not collect names, emails, or identity-linked information. Users in Kenya have rights under the DPA (including access, rectification, and erasure of personal data).
- General data protection principles — We follow data minimization, purpose limitation, and storage limitation principles regardless of jurisdiction.

If you believe your data protection rights have been violated, you may contact the Office of the Data Protection Commissioner (Kenya) or reach us directly.

10. Contact

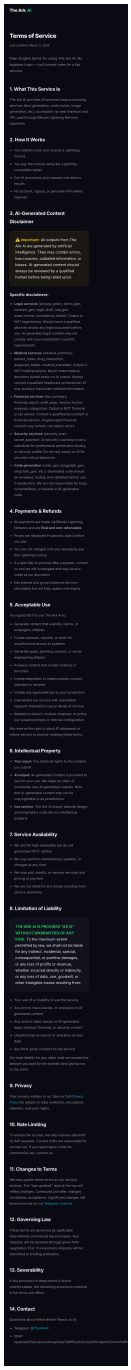
Questions about this policy? Reach us at:

- Telegram: @Yaadon
- Nostr: npub1ln0f7qmuxxlnzkcqzq3qe7k3f8fyvz2k5u5t8lcxzf4k2cnr5ldfm

3.10 Terms of Service

The Terms of Service page outlines the legal agreement between users and The Ark AI platform, covering acceptable use, payment terms, intellectual property, limitations of liability, and dispute resolution.

Figure: Terms of Service Page



The Ark AI

Terms of Service

Last updated: [illegible]

[illegible] AI, by [illegible] — [illegible] rules for a fair service.

1. What This Service Is

The Ark AI provides AI-powered text processing services (text generation, code review, image generation, etc.) accessible via web interface and API, paid through Bitcoin Lightning Network micropayments.

2. How It Works

- You submit a task and receive a Lightning invoice
- You pay the invoice using any Lightning-compatible wallet
- Our AI processes your request and returns results
- No account, signup, or personal information required

3. AI-Generated Content Disclaimer

> ⚠ **Important:** All outputs from The Ark AI are generated by artificial intelligence. They may contain errors, inaccuracies, outdated information, or biases. AI-generated content should always be reviewed by a qualified human before being relied upon.

Specific disclaimers:

- **Legal services:** [illegible] AI output is NOT legal advice. Always have a qualified attorney review any legal document before use. [illegible] content may not comply with your jurisdiction's specific requirements.
- **Medical services:** [illegible] AI output is NOT medical advice. Never make medical decisions based solely on AI output. Always consult a qualified healthcare professional. AI may produce inaccurate medical information.
- **Financial services:** [illegible] AI output is NOT financial advice. [illegible] Consult a qualified accountant or financial advisor. AI-generated financial content may contain calculation errors.
- **Security services:** [illegible] AI security scanning is not a substitute for professional penetration testing. [illegible] AI may miss critical vulnerabilities or flag false positives.
- **Code generation:** [illegible] AI-generated code should be reviewed, tested, and validated before use in production. We are not responsible for bugs, vulnerabilities, or failures in AI-generated code.

4. Payments & Refunds

- All payments are made via Bitcoin Lightning Network and are **final and non-refundable**
- Prices are displayed in satoshis (sats) before you pay
- You are not charged until you explicitly pay the Lightning invoice
- If a task fails to execute after payment, contact us and we will investigate and may issue a credit at our discretion
- Gift [illegible] and group balances are non-refundable but are fully usable until expiry

5. Acceptable Use

You agree NOT to use The Ark AI to:

- Generate content that exploits, harms, or endangers children
- Create malware, exploits, or tools for unauthorized access to systems
- Generate spam, phishing content, or social engineering attacks
- Produce content that incites violence or terrorism
- Create deepfakes or impersonation content intended to deceive
- Violate any applicable law in your jurisdiction
- Overwhelm our service with automated requests (intended to cause denial of service)
- Attempt to extract, reverse-engineer, or probe our system prompts or internal configuration

We reserve the right to block IP addresses or refuse service to anyone violating these terms.

6. Intellectual Property

- **Your input:** You retain all rights to the content you submit
- **Accepted** AI-generated content is provided to you under a CC0 (public domain) or equivalent license. AI-generated outputs may not be copyrightable in all jurisdictions.
- **Our services:** The Ark AI brand, website design, and proprietary code are our intellectual property

7. Service Availability

- We aim for high availability but do not guarantee 100% uptime
- We may perform maintenance, updates, or changes at any time
- We may add, modify, or remove services and pricing at any time
- We are not liable for any losses resulting from service downtime

8. Limitation of Liability

> **THE ARK AI IS PROVIDED "AS IS" WITHOUT WARRANTIES OF ANY KIND.** To the maximum extent permitted by law, we shall not be liable for any indirect, incidental, special, consequential, or punitive damages, or any loss of profits or revenue, whether incurred directly or indirectly, or any loss of data, use, goodwill, or other intangible losses resulting from:

- Your use of or inability to use the service
- Any errors, inaccuracies, or omissions in AI-generated content
- Any actions taken based on AI-generated legal, medical, financial, or security content
- Unauthorized access to or alteration of your data
- Any third-party conduct on the service

Our total liability for any claim shall not exceed the amount you paid for the specific task giving rise to the claim.

9. Privacy

Your privacy matters to us. See our Privacy Policy for full details on data collection, encryption, retention, and your rights.

10. Rate Limiting

To ensure fair access, we may impose rate limits on API requests. Current limits are reasonable for normal use. If you need higher limits for commercial use, contact us.

11. Changes to Terms

We may update these terms at any service version. The "last updated" date at the top will reflect changes. Continued use after changes constitutes acceptance. Significant changes will be announced via our Telegram channel.

12. Governing Law

These terms are governed by applicable international contract law and trader law. Any disputes will be resolved through good-faith negotiation first. If unsuccessful, disputes will be submitted to binding arbitration.

13. Severability

If any provision of these terms is found unenforceable, the remaining provisions continue in full force and effect.

14. Contact

Questions about these terms? Reach us at:

- Telegram: @thearkai
- Nostr: [illegible]

4. Key Source Code

The following section presents key excerpts from the original source code of The Ark AI platform. The complete software consists of approximately 3,500 lines of original Python source code across 12 modules, plus approximately 5,000 lines of HTML/CSS/JavaScript frontend code. API keys and secrets have been redacted for security.

4.1 Source Code Structure

- server.py (2,608 lines) - Core API server: task routing, streaming, payments, wallets, PDFs
- llm_tasks.py (746 lines) - AI task engine: 143+ services, prompt engineering, model routing
- pdf_gen.py (140 lines) - PDF generation: markdown-to-PDF conversion with branding
- wallet.py - Lightning wallet system with encrypted storage
- openai_compat.py - OpenAI-compatible API layer (/v1/chat/completions)
- workflows.py - Multi-step task orchestration engine
- config.yaml - Service pricing and model configuration
- index.html (~3,000 lines) - Main web interface with chat, payments, streaming
- translate.html - Dedicated translation interface
- groups.html - Chama group AI access interface
- about.html - Platform information and API documentation
- market.html - Services marketplace

server.py - Core API Server (first 100 lines)

```
"""
Bitcoin AI Agent Server
FastAPI server with L402 Lightning payment gating.

Usage:
    python server.py                    # uses config.yaml
    LNBITS_URL=... python server.py     # env overrides
"""

import os
import sys
sys.path.insert(0, os.path.join(os.path.dirname(__file__), "../data"))
from tracking import track_event, get_report
import sys
import time
import json
import hmac
import hashlib
import base64
import logging
import secrets
import sqlite3
import threading
from pathlib import Path
from typing import Any, Optional

import httpx
import yaml
from dotenv import load_dotenv
from fastapi import FastAPI, Request, Response, HTTPException
from fastapi.responses import JSONResponse
from fastapi.responses import FileResponse as _FileResp
from pydantic import BaseModel
from pdf_gen import generate_pdf, PDF_ELIGIBLE_TASKS, cleanup_old_pdfs, PDF_DIR

from workflows import WORKFLOWS
MAX_INPUT_LENGTH = 50000  # 50KB max input
MAX_TASK_INPUT = 100000  # 100KB for research tasks

from llm_tasks import execute_task as _execute_task_raw, estimate_price, TASK_TYPES, text_to_speech, generate_image, ...
from llm_tasks import get_model_for_task, FAST_MODEL, POWER_MODEL, POWER_TASKS
from openai_compat import register_openai_routes
from wallet import register_wallet_routes, wallet_deduct, wallet_balance

def _auto_refund(client_ip, amount_sats, payment_hash, reason="task_failure"):
    """Auto-credit a user when their paid task fails."""
    import json, time
    credits_file = "/opt/bitcoin-agent/data/credits.json"
    try:
        try:
            with open(credits_file, "r") as f:
                credits = json.load(f)
        except FileNotFoundError:
            credits = {}
        if client_ip in credits:
            credits[client_ip]["credit_sats"] += amount_sats
        else:
            credits[client_ip] = {
                "credit_sats": amount_sats,
                "reason": reason,
                "payment_hash": payment_hash,
                "created_at": time.time(),
        "client_ip": client_ip,
            }
        with open(credits_file, "w") as f:
            json.dump(credits, f, indent=2)
        logger.info(f"Auto-refund: {amount_sats} sats credited to {client_ip} (hash: {payment_hash[:16]}...)")
        return True
    except Exception as e:
        logger.error(f"Auto-refund failed: {e}")
        return False

_LEGAL_TASKS = {"contract_analysis", "law_research", "document_template", "nda_template", "regulatory_research"}
_MEDICAL_TASKS = {"health_lit_review", "clinical_notes_format", "pharma_research", "health_simplify", "health_research"}
_LEGAL_DISCLAIMER = "\n\n---\n?? **Disclaimer:** This is AI-generated general research content for informational and ...

import asyncio as _asyncio_locks
_task_locks: dict[str, _asyncio_locks.Lock] = {}

def _get_task_lock(payment_hash: str) -> _asyncio_locks.Lock:
    if payment_hash not in _task_locks:
        _task_locks[payment_hash] = _asyncio_locks.Lock()
    return _task_locks[payment_hash]

_MEDICAL_DISCLAIMER = "\n\n---\n?? **Disclaimer:** This is AI-generated general research content for informational an...

```

```
_FINANCIAL_TASKS = {"tax_summary", "invoice_format", "financial_report", "expense_categorize", "audit_prep"}
_FINANCIAL_DISCLAIMER = "\n\n---\n\u26a0\ufe0f **Disclaimer:** AI-generated content for informational purposes only. ...

_CODE_TASKS = {"code_gen", "code_review", "bug_finder", "unit_test", "security_scan", "secret_scanner", "dockerfile_g...
_CODE_DISCLAIMER = "\n\n---\n\u26a0\ufe0f **Disclaimer:** AI-generated code. Review and test thoroughly before deploy...

_SKIP_UNIVERSAL_DISCLAIMER = {"image_gen", "image_create", "image_edit", "image_variation", "image_upscale", "image_b...

# === Referral System ===
import json as _json_ref
import time as _time_ref
import random as _random_ref
import string as _string_ref

```

llm_tasks.py - AI Task Engine (first 100 lines)

```
"""
LLM Task Execution Module
Handles all AI task types with configurable LLM provider (OpenAI / Anthropic).
"""

import os
import logging
from typing import Any
from dataclasses import dataclass

# ---------------------------------------------------------------------------
# Model Tiering: fast model for cheap tasks, power model for expensive ones
# ---------------------------------------------------------------------------
FAST_MODEL = "Qwen/Qwen3-30B-A3B"
POWER_MODEL = "Qwen/Qwen3-235B-A22B-Instruct-2507"

POWER_TASKS = {
    "code_review", "code_gen", "bug_finder", "unit_test", "security_scan", "secret_scanner",
    "tech_blog", "content_write", "privacy_policy", "terms_gen", "nda_template", "document_template",
    "law_research", "contract_draft", "contract_gen", "contract_analysis", "lease_review",
    "business_plan", "pitch_deck", "market_research", "competitive_analysis",
    "health_lit_review", "health_research", "pharma_research", "clinical_notes_format",
    "financial_report", "tax_summary", "audit_prep",
    "lesson_plan", "curriculum_design", "study_guide",
    "property_analysis", "listing_write",
    "proposal_write", "proposal_draft", "sow_gen",
    "agent_design", "agent_prompt_eng", "agent_debug",
    "langchain_gen", "langgraph_gen", "multi_agent_gen",
    "data_analyze", "schema_design", "test_plan",
    "api_design", "api_doc", "k8s_gen", "terraform_gen",
    "seo_analysis", "competitor_analysis",
}

def get_model_for_task(task_type, default_model):
    """Return the appropriate model based on task complexity."""
    if task_type in POWER_TASKS:
        return POWER_MODEL
    return FAST_MODEL


logger = logging.getLogger(__name__)

@dataclass
class TaskResult:
    """Result from an LLM task execution."""
    content: str
    input_tokens: int
    output_tokens: int
    task_type: str
    model: str

    @property
    def total_tokens(self) -> int:
        return self.input_tokens + self.output_tokens


# --- System prompts per task type ---

ARK_IDENTITY = "You are The Ark AI ? a fast, reliable AI assistant powered by Bitcoin Lightning. NEVER identify yours...

SYSTEM_PROMPTS = {
    "summarize": "You are a concise summarizer. Provide a clear, accurate summary of the given text. Include key poin...
    "translate": "You are a professional translator. Translate the text accurately, preserving tone and meaning. If n...
    "code_gen": "You are an expert programmer. Generate clean, well-commented, production-quality code. Include brief...
    "research": "You are a thorough researcher. Provide well-structured, factual analysis with multiple perspectives....
    "content_write": "You are a skilled content writer. Write engaging, well-structured content tailored to the speci...
    "data_analyze": "You are a data analyst. Analyze the provided data, identify patterns, and present clear insights...
    "image_describe": "You are an image accessibility expert. Provide detailed, accurate descriptions of images for v...
    "code_review": "You are a principal engineer doing a thorough code review. Return a structured report with these ...
    "bug_finder": "You are a debugging expert. Analyze the code and identify potential bugs, edge cases, race conditi...
    "code_explain": "You are a code teacher. Explain what the code does in plain English, step by step. Make it under...
    "regex_gen": "You are a regex expert. Generate a regular expression that matches the described pattern. Include e...
    "sql_optimize": "You are a database performance expert. Optimize the SQL query for better performance. Explain wh...
    "doc_gen": "You are a professional document writer. Generate well-structured, comprehensive documents based on th...
    "commit_msg": "You are a git expert. Generate a clear, conventional commit message for the provided code diff. Us...
    "unit_test": "You are a senior QA engineer. Generate production-quality unit tests. For each function/method: ## ...
    "dockerfile_gen": "You are a Docker expert. Generate a production-ready Dockerfile for the described application....
    "cicd_gen": "You are a CI/CD expert. Generate a CI/CD pipeline configuration (GitHub Actions format) for the desc...
    "nginx_gen": "You are an Nginx configuration expert. Generate an Nginx config for the described routing requireme...
    "compose_gen": "You are a Docker Compose expert. Generate a docker-compose.yml for the described stack. Include n...
    "json_csv": "You are a data conversion expert. Convert the provided data between JSON and CSV formats. Preserve a...
    "data_clean": "You are a data cleaning expert. Clean the provided messy data: fix formatting, remove duplicates, ...
    "api_parse": "You are an API expert. Parse the provided API response and extract the requested fields. Return str...
    "cron_gen": "You are a cron expert. Generate a cron expression for the described schedule. Explain each field.",
    "readme_gen": "You are a README expert. Generate a comprehensive README.md for the described project. Include bad...
    "api_doc": "You are an API documentation expert. Generate comprehensive API documentation in OpenAPI/Swagger form...
```

```
    "tech_blog": "You are a technical blogger. Write an engaging, well-researched technical blog post on the given to...
    "changelog_gen": "You are a changelog expert. Generate a well-formatted changelog from the provided commit messag...
    "privacy_policy": "You are a legal document expert. Generate a comprehensive privacy policy for the described ser...
    "terms_gen": "You are a legal document expert. Generate terms of service for the described service. Include liabi...
    "seo_meta": "You are an SEO expert. Generate optimized meta descriptions, title tags, and OG tags for the describ...
    "ad_copy": "You are an advertising copywriter. Write compelling ad copy for the described product/service. Includ...
    "email_subject": "You are an email marketing expert. Generate 10 compelling email subject lines for the described...
    "color_palette": "You are a design expert. Generate a color palette for the described brand/project. Include hex ...

    "seo_analysis": "You are a senior SEO consultant performing a site audit. Return a structured report: ## Technica...
    "keyword_research": "You are a keyword research expert. Given a topic or niche, generate target keywords with sea...
    "competitor_analysis": "You are a competitive intelligence analyst. Analyze the described business and provide co...
    "grammar_check": "You are a professional editor. Return: 1) CORRECTED TEXT ? the clean version ready to use, 2) C...
    "rewrite": "You are a professional rewriter. Rewrite text to improve clarity, engagement, and flow while preservi...
```

pdf_gen.py - PDF Generation Module (complete)

```python
"""PDF generation module for document-type tasks."""
import os
import re
import uuid
import time
from fpdf import FPDF

PDF_DIR = "/opt/bitcoin-agent/data/pdfs"
os.makedirs(PDF_DIR, exist_ok=True)

PDF_ELIGIBLE_TASKS = {
    "doc_gen", "api_doc", "privacy_policy", "terms_gen", "nda_template",
    "document_template", "contract_gen", "contract_analysis", "contract_draft",
    "invoice_gen", "invoice_format", "business_plan", "proposal_write",
    "proposal_draft", "report_write", "sow_gen", "resume_gen", "cover_letter",
    "law_research", "lease_review", "memo_write", "whitepaper",
    "regulatory_research", "market_research", "competitive_analysis",
    "pitch_deck", "sop_gen",
}


def _safe(text):
    """Strip non-latin1 chars that Helvetica cannot render."""
    return text.encode("latin-1", "replace").decode("latin-1")


class ArkPDF(FPDF):
    def header(self):
        self.set_font("Helvetica", "B", 10)
        self.set_text_color(100, 100, 100)
        self.cell(0, 8, "The Ark AI", align="R", new_x="LMARGIN", new_y="NEXT")
        self.line(10, self.get_y(), 200, self.get_y())
        self.ln(4)

    def footer(self):
        self.set_y(-15)
        self.set_font("Helvetica", "I", 8)
        self.set_text_color(150, 150, 150)
        self.cell(0, 10, f"Generated by The Ark AI  |  Page {self.page_no()}/{26}", align="C")


def _clean(text):
    """Strip markdown inline formatting."""
    text = re.sub(r"\*\*(.+?)\*\*", r"\1", text)
    text = re.sub(r"\*(.+?)\*", r"\1", text)
    text = re.sub(r"`(.+?)`", r"\1", text)
    text = re.sub(r"\[(.+?)\]\(.+?\)", r"\1", text)
    return _safe(text)


def generate_pdf(content, task_type="document", title=None):
    pdf = ArkPDF()
    pdf.alias_nb_pages()
    pdf.add_page()
    pdf.set_auto_page_break(auto=True, margin=25)

    if title:
        pdf.set_font("Helvetica", "B", 20)
        pdf.set_text_color(20, 20, 20)
        pdf.set_x(10)
        pdf.multi_cell(0, 10, _safe(title), align="C")
        pdf.ln(4)

    label = task_type.replace("_", " ").title()
    pdf.set_font("Helvetica", "I", 10)
    pdf.set_text_color(120, 120, 120)
    pdf.set_x(10)
    pdf.cell(0, 6, _safe(label), align="C", new_x="LMARGIN", new_y="NEXT")
    pdf.ln(6)

    in_code = False
    for line in content.split("\n"):
        s = line.strip()

        if s.startswith("```"):
            in_code = not in_code
            continue

        if in_code:
            pdf.set_font("Courier", "", 9)
            pdf.set_text_color(50, 50, 50)
            pdf.set_fill_color(245, 245, 245)
            pdf.set_x(10)
            pdf.multi_cell(0, 5, _safe(line), fill=True)
            continue

```

```
        if s.startswith("##"):
            level = len(s) - len(s.lstrip("#"))
            text = s.lstrip("# ")
            size = max(10, 18 - level * 2)
            pdf.ln(3)
            pdf.set_font("Helvetica", "B", size)
            pdf.set_text_color(20, 20, 20)
            pdf.set_x(10)
            pdf.multi_cell(0, max(6, size * 0.6), _clean(text))
            pdf.ln(2)
        elif s.startswith("#"):
            text = s.lstrip("# ")
            pdf.ln(3)
            pdf.set_font("Helvetica", "B", 16)
            pdf.set_text_color(20, 20, 20)
            pdf.set_x(10)
            pdf.multi_cell(0, 9, _clean(text))
            pdf.ln(2)
        elif s.startswith("---") or s.startswith("***"):
            pdf.ln(2)
            pdf.line(10, pdf.get_y(), 200, pdf.get_y())
            pdf.ln(4)
        elif s.startswith("- ") or s.startswith("* "):
            pdf.set_font("Helvetica", "", 10)
            pdf.set_text_color(40, 40, 40)
            pdf.set_x(10)
            pdf.multi_cell(0, 7, "  - " + _clean(s[2:]))
        elif re.match(r"^\d+\.", s):
            pdf.set_font("Helvetica", "", 10)
            pdf.set_text_color(40, 40, 40)
            pdf.set_x(10)
            pdf.multi_cell(0, 6, "  " + _clean(s))
        elif s == "":
            pdf.ln(3)
        else:
            pdf.set_font("Helvetica", "", 10)
            pdf.set_text_color(40, 40, 40)
            pdf.set_x(10)
            pdf.multi_cell(0, 7, _clean(s))

    file_id = uuid.uuid4().hex[:12]
    filename = f"{task_type}_{file_id}.pdf"
    filepath = os.path.join(PDF_DIR, filename)
    pdf.output(filepath)
    return filename


def cleanup_old_pdfs(max_age_hours=24):
    now = time.time()
    cutoff = now - (max_age_hours * 3600)
    for f in os.listdir(PDF_DIR):
        fp = os.path.join(PDF_DIR, f)
        if os.path.getmtime(fp) < cutoff:
            os.remove(fp)
```

5. Declaration of Originality

I hereby declare that The Ark AI software described in this document is my original work. The source code, architecture, system design, prompt engineering, user interface design, and all creative elements described herein were conceived, designed, and written by me.

The software utilizes open-source libraries and frameworks (FastAPI, fpdf2, httpx) under their respective permissive licenses (MIT, Apache 2.0), and interfaces with third-party APIs (OpenAI, DeepInfra, LNbits) under their commercial terms of service. The original creative work subject to this copyright registration is the unique combination, architecture, implementation, user interface design, and service design built upon these foundations.

This document, the accompanying screenshots, and the source code excerpts constitute the complete submission for copyright registration of The Ark AI software platform under the Kenya Copyright Act, 2001 (Cap 130) as a Literary Work (Computer Program).

Signature: _____

Full Name: _____

ID Number: _____

Date: _____